

# Introduction to Artificial Intelligence

Study the official course objectives in the source syllabus.

OFFICIAL SOURCE DECLARATION

## How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 6342A is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

**Source file:** 6342A.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

## Official course information

Course information

| Item             | Official information                                  |
|------------------|-------------------------------------------------------|
| Programme        | Diploma in Artificial Intelligence & Machine Learning |
| Course code      | 6342A                                                 |
| Course title     | Introduction to Artificial Intelligence               |
| Semester         | 6 Credits: 4                                          |
| Course category  | Open Elective                                         |
| Periods per week | 4 (L:4 T:0 P:0) Periods per semester: 60              |

| Item                 | Official information                                 |
|----------------------|------------------------------------------------------|
| Periods per semester | 60                                                   |
| Credits              | 4                                                    |
| Prerequisite         | See the official syllabus PDF                        |
| Assessment           | Use the official assessment section reproduced below |

## Modules

4 official module sections detected.

## Topic entries

19 source-aligned topic entries retained.

## Study mode

Theory and application emphasis.

### STUDY METHOD

## How to use this lesson

### First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

### Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

### Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

## Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

### LEARNING OUTCOMES

## Objectives, outcomes and mapping

### Course objectives

- Study the official course objectives in the source syllabus.

#### Course outcomes

| CO  | Official outcome                                                                                       | Study level           |
|-----|--------------------------------------------------------------------------------------------------------|-----------------------|
| CO1 | Explain the concept of Artificial Intelligence 13 Understanding Apply Mathematical logic for knowledge | See official syllabus |
| CO2 | representation of AI 15 Understanding                                                                  | See official syllabus |
| CO3 | Demonstrate supervised machine learning techniques 15 Understanding                                    | See official syllabus |
| CO4 | Demonstrate Unsupervised Learning techniques 15 Understanding                                          | See official syllabus |

#### CO-PO mapping evidence

| Row | Extracted official mapping |
|-----|----------------------------|
| CO1 | CO1 3                      |
| CO2 | CO2 3                      |
| CO3 | CO3 3                      |
| CO4 | CO4 3                      |

### COVERAGE AUDIT

# Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

## Module coverage

| Module   | Official heading                                            | Topic entries | Hours        |
|----------|-------------------------------------------------------------|---------------|--------------|
| Module 1 | Explain the concept of Artificial Intelligence              | 5             | As specified |
| Module 2 | Apply Mathematical logic for knowledge representation of AI | 6             | As specified |
| Module 3 | Demonstrate supervised machine learning techniques          | 5             | As specified |
| Module 4 | Demonstrate Unsupervised Learning techniques                | 3             | As specified |

## MODULE 1

# Explain the concept of Artificial Intelligence

This module is presented from the official syllabus scope for Introduction to Artificial Intelligence. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Explain the concept of Artificial Intelligence
- Official duration: As specified
- Official topic entries captured: 5

- The full source excerpt is preserved below for coverage checking.

## Define AI, foundation and history of AI. 2 Understanding Understanding

Define AI, foundation and history of AI. 2 Understanding Understanding is an official topic in Module 1 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Define AI, foundation and history of AI. 2 Understanding Understanding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, define ai, foundation and history of ai. 2 understanding understanding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                  |
|---------------------|------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Define AI, foundation and history of AI. 2 Understanding Understanding means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                          |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 1- Define AI, foundation and history of AI. 2

Understanding Understanding definition/meaning

definition/meaning. definition/meaning, steps, application

definition/meaning. Technical terms English- definition/meaning.

## Discuss applications of AI. 2

Discuss applications of AI. 2 is an official topic in Module 1 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Discuss applications of AI. 2 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, discuss applications of ai. 2 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Discuss applications of AI. 2 means in the official course context           |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.



**Malayalam support:** Module 1- Discuss applications of AI. 2 definition/meaning application definition/meaning. application definition/meaning, steps, application definition/meaning application. Technical terms English- application definition/meaning.

## Explain the Turing test.

Explain the Turing test. is an official topic in Module 1 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Explain the Turing test. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, explain the turing test. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Explain the Turing test. means in the official course context                |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1- Explain the Turing test. definition/meaning . steps, application . Technical terms English-

## Explain AI problems

Explain AI problems is an official topic in Module 1 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Explain AI problems using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, explain ai problems should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Explain AI problems means in the official course context                     |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1- Explain AI problems definition/meaning . steps, application . Technical terms English-

**Explain agents and environments. 2 Understanding Contents : Introduction to AI-AI concepts, Foundations of AI, Applications of AI -Language models - Information retrieval - Information extraction - Natural language processing - Machine translation - Speech recognition, Turing test. AI Problems: Water Jug problem, Tower of Hanoi problem, Four-Queens problem, Travelling Salesman problem Intelligent Agents-Agents and Environments, Structure of Agent - types of agents.**

Explain agents and environments. 2 Understanding Contents : Introduction to AI-AI concepts, Foundations of AI, Applications of AI -Language models - Information retrieval - Information extraction - Natural language processing - Machine translation - Speech recognition, Turing test. AI Problems: Water Jug problem, Tower of Hanoi problem, Four-Queens problem, Travelling Salesman problem Intelligent Agents-Agents and Environments, Structure of Agent - types of agents. is an official topic in Module 1 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### **Study points**

- Define or identify Explain agents and environments. 2 Understanding Contents : Introduction to AI-AI concepts, Foundations of AI, Applications of AI -Language models - Information retrieval - Information extraction - Natural language processing - Machine translation - Speech recognition, Turing test. AI Problems: Water Jug problem, Tower of Hanoi problem, Four-Queens problem, Travelling Salesman problem Intelligent Agents-Agents and Environments, Structure of Agent - types of agents. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### **Engineering or professional relevance**

In Polytechnic practice, explain agents and environments. 2 understanding contents : introduction to ai-ai concepts, foundations of ai, applications of ai -language models -

information retrieval - information extraction - natural language processing - machine translation - speech recognition, turing test. ai problems: water jug problem, tower of hanoi problem, four-queens problem, travelling salesman problem intelligent agents- agents and environments, structure of agent - types of agents. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Explain agents and environments. 2 Understanding Contents : Introduction to AI-AI concepts, Foundations of AI, Applications of AI -Language models - Information retrieval - Information extraction - Natural language processing - Machine translation - Speech recognition, Turing test. AI Problems: Water Jug problem, Tower of Hanoi problem, Four-Queens problem, Travelling Salesman problem Intelligent Agents-Agents and Environments, Structure of Agent - types of agents. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                               |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-എന്നു പറയുന്ന Explain agents and environments. 2 Understanding Contents : Introduction to AI-AI concepts, Foundations of AI, Applications of AI -Language models - Information retrieval - Information extraction - Natural language processing - Machine translation - Speech recognition, Turing test. AI Problems: Water Jug problem, Tower of Hanoi problem, Four-Queens problem, Travelling Salesman problem Intelligent Agents-Agents and Environments, Structure of Agent - types of agents. എന്നാൽ definition/meaning എന്നാണ്. എന്നാണ് steps, application എന്നാണ്. Technical terms English-ൽ പറയുന്ന എന്നാണ്.

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.  
Educational explanations below are separate elaboration.

Explain the concept of Artificial Intelligence

|                                                                                                                                         |                                          |   |
|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|---|
| M1.01                                                                                                                                   | Define AI, foundation and history of AI. | 2 |
| Understanding                                                                                                                           |                                          |   |
| Understanding                                                                                                                           |                                          |   |
| M1.02                                                                                                                                   | Discuss applications of AI.              | 2 |
| M1.03                                                                                                                                   | Explain the Turing test.                 | 3 |
| Understanding                                                                                                                           |                                          |   |
| M1.04                                                                                                                                   | Explain AI problems                      | 3 |
| Understanding                                                                                                                           |                                          |   |
| M1.05                                                                                                                                   | Explain agents and environments.         | 2 |
| Understanding                                                                                                                           |                                          |   |
| Contents :                                                                                                                              |                                          |   |
| Introduction to AI-AI concepts, Foundations of AI, Applications of AI -Language models                                                  |                                          |   |
| - Information retrieval - Information extraction - Natural language processing - Machine translation - Speech recognition, Turing test. |                                          |   |
| AI Problems: Water Jug problem, Tower of Hanoi problem, Four-Queens problem, Travelling Salesman problem                                |                                          |   |
| Intelligent Agents-Agents and Environments, Structure of Agent - types of agents.                                                       |                                          |   |

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

## MODULE 2

# Apply Mathematical logic for knowledge representation of AI

This module is presented from the official syllabus scope for Introduction to Artificial Intelligence. Use the topic cards for learning and the source transcript for exact coverage.



**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Apply Mathematical logic for knowledge representation of AI
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

## 2 Understanding knowledge representation and reasoning Describe the representation and reasoning

2 Understanding knowledge representation and reasoning Describe the representation and reasoning is an official topic in Module 2 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding knowledge representation and reasoning Describe the representation and reasoning using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding knowledge representation and reasoning describe the representation and reasoning should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 2 Understanding knowledge representation and reasoning Describe the representation and reasoning means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                    |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                          |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-പാഠ്യ 2 Understanding knowledge representation and reasoning Describe the representation and reasoning പരസ്പരം ബന്ധിതമായ പദങ്ങൾ definition/meaning പദത്തിന്റെ അർത്ഥം. പദങ്ങൾ പരസ്പരം ബന്ധിതമായ, steps, application പദങ്ങൾ പരസ്പരം ബന്ധിതമായ പദങ്ങൾ. Technical terms English-പദങ്ങൾ പരസ്പരം ബന്ധിതമായ പദങ്ങൾ.

## about objects, relations, events, actions, time 2 Understanding and space

about objects, relations, events, actions, time 2 Understanding and space is an official topic in Module 2 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify about objects, relations, events, actions, time 2 Understanding and space using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, about objects, relations, events, actions, time 2 understanding and space should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                     |
|---------------------|---------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What about objects, relations, events, actions, time 2 Understanding and space means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                             |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                   |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 2-പ്രകൃതി വസ്തുക്കൾ, ബന്ധങ്ങൾ, സംഭവങ്ങൾ, പ്രവൃത്തികൾ, സമയം 2  
Understanding and space പദപരിചയം definition/meaning പദപരിചയം.  
പദപരിചയം പദപരിചയം, steps, application പദപരിചയം പദപരിചയം പദപരിചയം. Technical  
terms English-പദപരിചയം പദപരിചയം.

## Demonstrate Propositional Logic

Demonstrate Propositional Logic is an official topic in Module 2 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Demonstrate Propositional Logic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, demonstrate propositional logic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Demonstrate Propositional Logic means in the official course context         |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2- Demonstrate Propositional Logic  $\wedge, \vee, \neg, \rightarrow, \leftrightarrow$  definition/meaning  $\wedge, \vee, \neg, \rightarrow, \leftrightarrow$ .  $\wedge, \vee, \neg, \rightarrow, \leftrightarrow$   $\wedge, \vee, \neg, \rightarrow, \leftrightarrow$ , steps, application  $\wedge, \vee, \neg, \rightarrow, \leftrightarrow$   $\wedge, \vee, \neg, \rightarrow, \leftrightarrow$ . Technical terms English-  $\wedge, \vee, \neg, \rightarrow, \leftrightarrow$   $\wedge, \vee, \neg, \rightarrow, \leftrightarrow$ .

## Demonstrate First Order Logic

Demonstrate First Order Logic is an official topic in Module 2 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Demonstrate First Order Logic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, demonstrate first order logic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Demonstrate First Order Logic means in the official course context           |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.



**Malayalam support:** Module 2- Demonstrate First Order Logic  $\forall$   $\exists$  definition/meaning  $\forall$   $\exists$ .  $\forall$   $\exists$   $\forall$   $\exists$ , steps, application  $\forall$   $\exists$   $\forall$   $\exists$ . Technical terms English-  $\forall$   $\exists$   $\forall$   $\exists$ .

## Explain Soundness and Completeness

Explain Soundness and Completeness is an official topic in Module 2 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Explain Soundness and Completeness using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, explain soundness and completeness should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Explain Soundness and Completeness means in the official course context      |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-എന്നു വിശദീകരിക്കുക ശബ്ദതയ്ക്കും പൂർണ്ണതയ്ക്കും

എന്നിവയുടെ നിർവ്വചനം/അർത്ഥം എഴുതുക. നിർവ്വചനം നിർവ്വചനം, steps, application എന്നിവയുടെ നിർവ്വചനം എഴുതുക. Technical terms English-ൽ എഴുതുക എഴുതുക.

**Explain Forward and Backward chaining 3 Understanding Knowledge representation and reasoning-Ontologies,foundations of knowledge representation and reasoning, representing and reasoning about objects,relations,events,actions, time and space. Logic and Inferences: Propositional Logic, First Order Logic, Soundness and Completeness, Forward and Backward chaining.**

Explain Forward and Backward chaining 3 Understanding Knowledge representation and reasoning-Ontologies,foundations of knowledge representation and reasoning, representing and reasoning about objects,relations,events,actions, time and space. Logic and Inferences: Propositional Logic, First Order Logic, Soundness and Completeness, Forward and Backward chaining. is an official topic in Module 2 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### **Study points**

- Define or identify Explain Forward and Backward chaining 3 Understanding Knowledge representation and reasoning-Ontologies,foundations of knowledge representation and reasoning, representing and reasoning about objects,relations,events,actions, time and space. Logic and Inferences: Propositional Logic, First Order Logic, Soundness and Completeness, Forward and Backward chaining. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### **Engineering or professional relevance**

In Polytechnic practice, explain forward and backward chaining 3 understanding knowledge representation and reasoning-ontologies,foundations of knowledge representation and reasoning, representing and reasoning about objects,relations,events,actions, time and space. logic and inferences: propositional logic, first order logic, soundness and completeness, forward and backward chaining.

should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Explain Forward and Backward chaining 3 Understanding Knowledge representation and reasoning-Ontologies,foundations of knowledge representation and reasoning, representing and reasoning about objects,relations,events,actions, time and space. Logic and Inferences: Propositional Logic, First Order Logic, Soundness and Completeness, Forward and Backward chaining. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                              |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                    |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2- Explain Forward and Backward chaining 3 Understanding Knowledge representation and reasoning-Ontologies,foundations of knowledge representation and reasoning, representing and reasoning about objects,relations,events,actions, time and space. Logic and Inferences: Propositional Logic, First Order Logic, Soundness and Completeness, Forward and Backward chaining. definition/meaning . steps, application . Technical terms English- .

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.  
Educational explanations below are separate elaboration.

Apply Mathematical logic for knowledge representation of AI

|                                                                                                                                                                                                 |                                                                                                     |   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|---|
| M2.01                                                                                                                                                                                           | Summarize Ontologies, foundations of knowledge representation and reasoning                         | 2 |
| Understanding                                                                                                                                                                                   |                                                                                                     |   |
| M2.02                                                                                                                                                                                           | Describe the representation and reasoning about objects, relations, events, actions, time and space | 2 |
| Understanding                                                                                                                                                                                   |                                                                                                     |   |
| M2.03                                                                                                                                                                                           | Demonstrate Propositional Logic                                                                     | 3 |
| Understanding                                                                                                                                                                                   |                                                                                                     |   |
| M2.04                                                                                                                                                                                           | Demonstrate First Order Logic                                                                       | 2 |
| Understanding                                                                                                                                                                                   |                                                                                                     |   |
| M2.05                                                                                                                                                                                           | Explain Soundness and Completeness                                                                  | 3 |
| Understanding                                                                                                                                                                                   |                                                                                                     |   |
| M2.06                                                                                                                                                                                           | Explain Forward and Backward chaining                                                               | 3 |
| Understanding                                                                                                                                                                                   |                                                                                                     |   |
| Contents:                                                                                                                                                                                       |                                                                                                     |   |
| Knowledge representation and reasoning-Ontologies, foundations of knowledge representation and reasoning, representing and reasoning about objects, relations, events, actions, time and space. |                                                                                                     |   |
| Logic and Inferences: Propositional Logic, First Order Logic, Soundness and                                                                                                                     |                                                                                                     |   |

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

### MODULE 3

## Demonstrate supervised machine learning techniques

This module is presented from the official syllabus scope for Introduction to Artificial Intelligence. Use the topic cards for learning and the source transcript for exact

coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Demonstrate supervised machine learning techniques
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

## Explain the Machine Learning concepts. 3 Understanding Describe the process involved in machine

Explain the Machine Learning concepts. 3 Understanding Describe the process involved in machine is an official topic in Module 3 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Explain the Machine Learning concepts. 3 Understanding Describe the process involved in machine using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, explain the machine learning concepts. 3 understanding describe the process involved in machine should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                           |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Explain the Machine Learning concepts. 3 Understanding Describe the process involved in machine means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                   |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                         |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.



**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-എന്നു വിശദീകരിക്കുക. 3 Understanding Describe the process involved in machine learning. definition/meaning എന്നിവ വിശദീകരിക്കുക. എന്നിവയുടെ steps, application എന്നിവ വിശദീകരിക്കുക. Technical terms English-ൽ എഴുതുക.

### 3 Understanding learning techniques.

3 Understanding learning techniques. is an official topic in Module 3 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

#### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

#### Study points

- Define or identify 3 Understanding learning techniques. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

#### Engineering or professional relevance

In Polytechnic practice, 3 understanding learning techniques. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What 3 Understanding learning techniques. means in the official course context    |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-3 Understanding learning techniques.

പഠനരീതികൾ പറ്റി definition/meaning എഴുതുക. പഠനരീതികൾ, steps, application എഴുതുക. Technical terms English-ൽ എഴുതുക.

## Describe Supervised Learning

Describe Supervised Learning is an official topic in Module 3 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Describe Supervised Learning using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, describe supervised learning should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Describe Supervised Learning means in the official course context            |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3- Describe Supervised Learning   
 definition/meaning .   
 application   
 Technical terms English-   
 .

## Demonstrate Classification techniques

Demonstrate Classification techniques is an official topic in Module 3 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Demonstrate Classification techniques using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, demonstrate classification techniques should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Demonstrate Classification techniques means in the official course context   |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3- Demonstrate Classification techniques

ৱিকিপিডিয়াৰ পৰা definition/meaning গ্ৰহণ কৰা হৈছে. ৱিকিপিডিয়াৰ পৰা steps, application গ্ৰহণ কৰা হৈছে. Technical terms English-ৰ পৰা গ্ৰহণ কৰা হৈছে.

**Demonstrate Regression techniques 3 Understanding Machine Learning - Definition -Applications - Processes involved in Machine Learning, Machine Learning Techniques-Supervised Learning, Unsupervised Learning and Reinforcement Learning, Data preprocessing - Normalization, Missing value handling. Supervised Learning Classification - Learning a Class from Examples, Logistic Regression - K-Nearest Neighbours - Support Vector Machine - Naive Bayes - Decision Tree - Random Forest Regression- Linear Regression -Multi Linear Regression**

Demonstrate Regression techniques 3 Understanding Machine Learning - Definition -Applications - Processes involved in Machine Learning, Machine Learning Techniques-Supervised Learning, Unsupervised Learning and Reinforcement Learning, Data preprocessing - Normalization, Missing value handling. Supervised Learning Classification - Learning a Class from Examples, Logistic Regression - K-Nearest Neighbours - Support Vector Machine - Naive Bayes - Decision Tree - Random Forest Regression- Linear Regression -Multi Linear Regression is an official topic in Module 3 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

**Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

**Study points**

- Define or identify Demonstrate Regression techniques 3 Understanding Machine Learning - Definition -Applications - Processes involved in Machine Learning, Machine Learning Techniques-Supervised Learning, Unsupervised Learning and Reinforcement Learning, Data preprocessing - Normalization, Missing value handling. Supervised Learning Classification - Learning a Class from Examples, Logistic Regression - K-Nearest Neighbours - Support Vector Machine - Naive Bayes - Decision Tree - Random Forest Regression- Linear Regression -Multi Linear Regression using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.



## Engineering or professional relevance

In Polytechnic practice, demonstrate regression techniques 3 understanding machine learning - definition -applications - processes involved in machine learning, machine learning techniques-supervised learning, unsupervised learning and reinforcement learning, data preprocessing – normalization, missing value handling. supervised learning classification - learning a class from examples, logistic regression - k-nearest neighbours - support vector machine - naive bayes - decision tree - random forest regression- linear regression -multi linear regression should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Demonstrate Regression techniques 3 Understanding Machine Learning - Definition - Applications - Processes involved in Machine Learning, Machine Learning Techniques-Supervised Learning, Unsupervised Learning and Reinforcement Learning, Data preprocessing – Normalization, Missing value handling. Supervised Learning Classification - Learning a Class from Examples, Logistic Regression - K-Nearest Neighbours - Support Vector Machine - Naïve Bayes - Decision Tree - Random Forest Regression- Linear Regression -Multi Linear Regression means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3- Demonstrate Regression techniques 3 Understanding Machine Learning - Definition -Applications - Processes involved in Machine Learning, Machine Learning Techniques-Supervised Learning, Unsupervised Learning and Reinforcement Learning, Data preprocessing – Normalization, Missing value handling. Supervised Learning Classification - Learning a Class from Examples, Logistic Regression - K-Nearest Neighbours - Support Vector Machine - Naive Bayes - Decision Tree - Random Forest Regression- Linear Regression -Multi Linear Regression definition/meaning . steps, application . Technical terms English- .

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.  
Educational explanations below are separate elaboration.

Demonstrate supervised machine learning techniques

|               |                                          |   |
|---------------|------------------------------------------|---|
| M3.01         | Explain the Machine Learning concepts.   | 3 |
| Understanding | Describe the process involved in machine |   |
| M3.02         | learning techniques.                     | 3 |
| Understanding |                                          |   |
| M3.03         | Describe Supervised Learning             | 3 |
| Understanding |                                          |   |
| M3.04         | Demonstrate Classification techniques    | 3 |
| Understanding |                                          |   |
| M3.05         | Demonstrate Regression techniques        | 3 |
| Understanding |                                          |   |

Contents:

Machine Learning - Definition -Applications - Processes involved in Machine Learning,

Machine Learning Techniques-Supervised Learning, Unsupervised Learning and Reinforcement Learning, Data preprocessing – Normalization, Missing value handling.

Supervised Learning

Classification - Learning a Class from Examples, Logistic Regression - K-Nearest Neighbours - Support Vector Machine - Naive Bayes - Decision Tree - Random Forest

Regression- Linear Regression -Multi Linear Regression

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

### MODULE 4

## Demonstrate Unsupervised Learning techniques

This module is presented from the official syllabus scope for Introduction to Artificial Intelligence. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Demonstrate Unsupervised Learning techniques
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

## Describe types of Clustering

Describe types of Clustering is an official topic in Module 4 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Describe types of Clustering using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, describe types of clustering should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Describe types of Clustering means in the official course context            |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- Describe types of Clustering  
definition/meaning . steps, application . Technical terms English-  
.

## Demonstrate Clustering techniques

Demonstrate Clustering techniques is an official topic in Module 4 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Demonstrate Clustering techniques using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, demonstrate clustering techniques should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Demonstrate Clustering techniques means in the official course context       |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- Demonstrate Clustering techniques

1. **Definition/Meaning:** The process of identifying and describing the steps, application, and technical terms associated with a specific task or process.

**Illustrate Dimensionality Reduction techniques 6 Understanding Clustering - Discovering clusters - Types of Clustering: Hierarchical, Agglomerative Clustering, Divisive clustering, Partitional Clustering - K-means clustering, Mean Shift clustering Dimensionality reduction - Principal Component Analysis, Factor Analysis, Multidimensional scaling, Linear Discriminant Analysis. Text / Reference T/R Book Title/Author Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd T1 Edition. Prentice Hall. T2 Nilsson N.J., Artificial Intelligence - A New Synthesis, Harcourt Asia Pvt. Ltd. Ethem Alpaydin, Introduction to Machine Learning, 2nd edition, MIT Press 2010. T3 T4 Tom Mitchell, Machine Learning, McGraw-Hill, 1997 R1 Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education R2 (India) Online Resources Sl.No Website Link 1 [https://onlinecourses.swayam2.ac.in/aic20\\_sp06/preview](https://onlinecourses.swayam2.ac.in/aic20_sp06/preview) 2 Introduction to Machine Learning - Course (nptel.ac.in)**

Illustrate Dimensionality Reduction techniques 6 Understanding Clustering - Discovering clusters - Types of Clustering: Hierarchical, Agglomerative Clustering, Divisive clustering, Partitional Clustering - K-means clustering, Mean Shift clustering Dimensionality reduction - Principal Component Analysis, Factor Analysis, Multidimensional scaling, Linear Discriminant Analysis. Text / Reference T/R Book Title/Author Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd T1 Edition. Prentice Hall. T2 Nilsson N.J., Artificial Intelligence - A New Synthesis, Harcourt Asia Pvt. Ltd. Ethem Alpaydin, Introduction to Machine Learning, 2nd edition, MIT Press 2010. T3 T4 Tom Mitchell, Machine Learning, McGraw-Hill, 1997 R1 Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education R2 (India) Online Resources Sl.No Website Link 1 [https://onlinecourses.swayam2.ac.in/aic20\\_sp06/preview](https://onlinecourses.swayam2.ac.in/aic20_sp06/preview) 2 Introduction to Machine Learning - Course (nptel.ac.in) is an official topic in Module 4 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### **Study points**



- Define or identify Illustrate Dimensionality Reduction techniques 6 Understanding Clustering - Discovering clusters - Types of Clustering: Hierarchical, Agglomerative Clustering, Divisive clustering, Partitional Clustering - K-means clustering, Mean Shift clustering Dimensionality reduction – Principal Component Analysis, Factor Analysis, Multidimensional scaling, Linear Discriminant Analysis. Text / Reference T/R Book Title/Author Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd T1 Edition. Prentice Hall. T2 Nilsson N.J., Artificial Intelligence - A New Synthesis, Harcourt Asia Pvt. Ltd. Ethem Alpaydin, Introduction to Machine Learning, 2nd edition, MIT Press 2010. T3 T4 Tom Mitchell, Machine Learning, McGraw-Hill, 1997 R1 Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education R2 (India) Online Resources Sl.No Website Link 1 [https://onlinecourses.swayam2.ac.in/aic20\\_sp06/preview](https://onlinecourses.swayam2.ac.in/aic20_sp06/preview) 2 Introduction to Machine Learning - Course (nptel.ac.in) using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, illustrate dimensionality reduction techniques 6 understanding clustering - discovering clusters - types of clustering: hierarchical, agglomerative clustering, divisive clustering, partitional clustering - k-means clustering, mean shift clustering dimensionality reduction – principal component analysis, factor analysis, multidimensional scaling, linear discriminant analysis. text / reference t/r book title/author stuart russell and peter norvig. artificial intelligence: a modern approach, 3rd t1 edition. prentice hall. t2 nilsson n.j., artificial intelligence - a new synthesis, harcourt asia pvt. ltd. ethem alpaydin, introduction to machine learning, 2nd edition, mit press 2010. t3 t4 tom mitchell, machine learning, mcgraw-hill, 1997 r1 elaine rich and kevin knight. artificial intelligence, tata mcgraw hill deepak khemani. a first course in artificial intelligence, mcgraw hill education r2 (india) online resources sl.no website link 1 [https://onlinecourses.swayam2.ac.in/aic20\\_sp06/preview](https://onlinecourses.swayam2.ac.in/aic20_sp06/preview) 2 introduction to machine learning - course (nptel.ac.in) should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Illustrate Dimensionality Reduction techniques 6 Understanding Clustering - Discovering clusters - Types of Clustering: Hierarchical, Agglomerative Clustering, Divisive clustering, Partitional Clustering - K-means clustering, Mean Shift clustering Dimensionality reduction – Principal Component Analysis, Factor Analysis, Multidimensional scaling, Linear Discriminant Analysis. Text / Reference T/R Book Title/Author Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd T1 Edition. Prentice Hall. T2 Nilsson N.J., Artificial |

| Aspect        | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | Intelligence - A New Synthesis, Harcourt Asia Pvt. Ltd. Ethem Alpaydin, Introduction to Machine Learning, 2nd edition, MIT Press 2010. T3 T4 Tom Mitchell, Machine Learning, McGraw-Hill, 1997 R1 Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education R2 (India) Online Resources Sl.No Website Link 1 <a href="https://onlinecourses.swayam2.ac.in/aic20_sp06/preview">https://onlinecourses.swayam2.ac.in/aic20_sp06/preview</a> 2 Introduction to Machine Learning - Course (nptel.ac.in) means in the official course context |
| Study lens    | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ Illustrate Dimensionality Reduction techniques 6 Understanding Clustering - Discovering clusters - Types of Clustering: Hierarchical, Agglomerative Clustering, Divisive clustering, Partitional Clustering - K-means clustering, Mean Shift clustering Dimensionality reduction – Principal Component Analysis, Factor Analysis, Multidimensional scaling, Linear Discriminant Analysis. Text / Reference T/R Book Title/Author Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd T1 Edition. Prentice Hall. T2 Nilsson N.J., Artificial Intelligence - A New Synthesis, Harcourt Asia Pvt. Ltd. Ethem Alpaydin, Introduction to Machine Learning, 2nd edition, MIT Press 2010. T3 T4 Tom Mitchell, Machine Learning, McGraw-Hill, 1997 R1 Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education R2 (India) Online Resources Sl.No Website Link 1 [https://onlinecourses.swayam2.ac.in/aic20\\_sp06/preview](https://onlinecourses.swayam2.ac.in/aic20_sp06/preview) 2 Introduction to Machine Learning - Course (nptel.ac.in) □□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□. □□□□□ □□□□□ □□□□□□, steps, application □□□□□ □□□□□□□□ □□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□.

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Demonstrate Unsupervised Learning techniques

|       |                                            |   |
|-------|--------------------------------------------|---|
| M4.01 | Describe types of Clustering Understanding | 3 |
|-------|--------------------------------------------|---|

|       |                                                 |   |
|-------|-------------------------------------------------|---|
| M4.02 | Demonstrate Clustering techniques Understanding | 6 |
|-------|-------------------------------------------------|---|

|       |                                                                        |   |
|-------|------------------------------------------------------------------------|---|
| M4.03 | Illustrate Dimensionality Reduction techniques Understanding Contents: | 6 |
|-------|------------------------------------------------------------------------|---|

Clustering - Discovering clusters - Types of Clustering: Hierarchical, Agglomerative Clustering, Divisive clustering, Partitional Clustering - K-means clustering, Mean Shift clustering

|                                                                                                                                                    |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Dimensionality reduction – Principal Component Analysis, Factor Analysis, Multidimensional scaling, Linear Discriminant Analysis. Text / Reference |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------|--|

|                                                                                  |                   |
|----------------------------------------------------------------------------------|-------------------|
| T/R                                                                              | Book Title/Author |
| Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd |                   |
| T1                                                                               |                   |

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

## RAPID REFERENCE

## Concept and formula bank

### Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

### Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

## RAPID REVISION

### Diagram and source-transcript library

#### Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

#### Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

#### Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

#### Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

## OFFICIAL SELF-LEARNING

# Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

## EXAMINATION PREPARATION

### Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

**Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

## ACTIVE RECALL

### Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

#### Module 1

**Define or explain Define AI, foundation and history of AI. 2 Understanding Understanding. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Define AI, foundation and history of AI. 2 Understanding Understanding*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Discuss applications of AI. 2. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Discuss applications of AI. 2.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Explain the Turing test.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Explain the Turing test..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Explain AI problems. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Explain AI problems.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain 2 Understanding knowledge representation and reasoning**  
**Describe the representation and reasoning. (3 marks)**

**Answer:** Begin with the standard definition or identity of *2 Understanding knowledge representation and reasoning*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain about objects, relations, events, actions, time 2 Understanding and space. (3 marks)**

**Answer:** Begin with the standard definition or identity of *about objects, relations, events, actions, time 2 Understanding and space*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Demonstrate Propositional Logic. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Demonstrate Propositional Logic*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Demonstrate First Order Logic. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Demonstrate First Order Logic*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

## Module 3

**Define or explain Explain the Machine Learning concepts. 3 Understanding Describe the process involved in machine. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Explain the Machine Learning concepts. 3 Understanding Describe the process involved in machine*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain 3 Understanding learning techniques.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *3 Understanding learning techniques..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Describe Supervised Learning. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Describe Supervised Learning*. State the governing principle, list the key components or stages, and close with one



relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

### Define or explain Demonstrate Classification techniques. (3 marks)

**Answer:** Begin with the standard definition or identity of *Demonstrate Classification techniques*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

## Module 4

### Define or explain Describe types of Clustering. (3 marks)

**Answer:** Begin with the standard definition or identity of *Describe types of Clustering*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

### Define or explain Demonstrate Clustering techniques. (3 marks)

**Answer:** Begin with the standard definition or identity of *Demonstrate Clustering techniques*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps,

**Define or explain Illustrate Dimensionality Reduction techniques 6 Understanding Clustering - Discovering clusters - Types of Clustering: Hierarchical, Agglomerative Clustering, Divisive clustering, Partitional Clustering - K-means clustering, Mean Shift clustering Dimensionality reduction - Principal Component Analysis, Factor Analysis, Multidimensional scaling, Linear Discriminant Analysis. Text / Reference T/R Book Title/Author Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd T1 Edition. Prentice Hall. T2 Nilsson N.J., Artificial Intelligence - A New Synthesis, Harcourt Asia Pvt. Ltd. Ethem Alpaydin, Introduction to Machine Learning, 2nd edition, MIT Press 2010. T3 T4 Tom Mitchell, Machine Learning, McGraw-Hill, 1997 R1 Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education R2 (India) Online Resources Sl.No Website Link 1 [https://onlinecourses.swayam2.ac.in/aic20\\_sp06/preview](https://onlinecourses.swayam2.ac.in/aic20_sp06/preview) 2 Introduction to Machine Learning - Course (nptel.ac.in). (3 marks)**

**Answer:** Begin with the standard definition or identity of *Illustrate Dimensionality Reduction techniques 6 Understanding Clustering - Discovering clusters - Types of Clustering: Hierarchical, Agglomerative Clustering, Divisive clustering, Partitional Clustering - K-means clustering, Mean Shift clustering Dimensionality reduction - Principal Component Analysis, Factor Analysis, Multidimensional scaling, Linear Discriminant Analysis. Text / Reference T/R Book Title/Author Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd T1 Edition. Prentice Hall. T2 Nilsson N.J., Artificial Intelligence - A New Synthesis, Harcourt Asia Pvt. Ltd. Ethem Alpaydin, Introduction to Machine Learning, 2nd edition, MIT Press 2010. T3 T4 Tom Mitchell, Machine Learning, McGraw-Hill, 1997 R1 Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education R2 (India) Online Resources Sl.No Website Link 1 [https://onlinecourses.swayam2.ac.in/aic20\\_sp06/preview](https://onlinecourses.swayam2.ac.in/aic20_sp06/preview) 2 Introduction to Machine Learning - Course (nptel.ac.in).* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□ □□□□□.

## Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

### Module 1

1-mark definition: State the meaning of **Define AI, foundation and history of AI. 2 Understanding Understanding.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 2

1-mark definition: State the meaning of **2 Understanding knowledge representation and reasoning Describe the representation and reasoning.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 3

1-mark definition: State the meaning of **Explain the Machine Learning concepts. 3 Understanding Describe the process involved in machine.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 4

1-mark definition: State the meaning of **Describe types of Clustering**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### QUICK REVISION

## Exam checklist

### Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

### Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

### REFERENCE VOCABULARY

## Glossary

### Study glossary

| Term           | Meaning in this lesson                                                   |
|----------------|--------------------------------------------------------------------------|
| Course Outcome | A capability the student should demonstrate after completing the course. |
| Module         | An official syllabus unit with defined topics and hours.                 |

| Term              | Meaning in this lesson                                                   |
|-------------------|--------------------------------------------------------------------------|
| CIE               | Continuous Internal Evaluation.                                          |
| ESE               | End Semester Examination.                                                |
| Source transcript | A preserved extract of the official syllabus used for coverage checking. |

## SOURCE-SUPPORTED REFERENCES

### References and web resources

#### References listed in the official PDF

References are included in the official source PDF.

#### Web resources listed in the official PDF

- Sl.No Website Link
- [https://onlinecourses.swayam2.ac.in/aic20\\_sp06/preview](https://onlinecourses.swayam2.ac.in/aic20_sp06/preview)
- Introduction to Machine Learning - Course (nptel.ac.in)

## IN-PAGE SEARCH

### Search this lesson

Prepared from the supplied official SITTTT syllabus PDF for Course 6342A. Verify institutional updates against the current official curriculum.